



CS503: Advance in Algorithms

Mid Semester, Autumn 2019

IIT Patna

Attempt all questions. Do not write anything on the question paper.

Time: 2 Hrs

Full marks: 30

1. For each pair of functions $f(n)$ and $g(n)$ in the table below, write O , Ω , or Θ in the appropriate space, depending on whether $f(n) = O(g(n))$, $f(n) = \Omega(g(n))$, or $f(n) = \Theta(g(n))$. If there is more than one relation between $f(n)$ and $g(n)$, write only the strongest one. The first line is a demo solution. We use $\lg$ to denote the base-2 logarithm. (3)

$f(n) \backslash g(n)$	n	$n \lg n$	n^2
$n \lg^2 n$	Ω	Ω	O
$2^{\lg^2 n}$			
$\lg(n!)$			
$n^{\lg 3}$			

2. You are given two sorted lists of size m and n . Give an $O(\lg m + \lg n)$ time algorithm for computing the k th smallest element in the union of the two lists. (5)
3. Consider the following *max-finding* algorithm for a set of n non-identical integers (X)
- M1: $j \leftarrow n, k \leftarrow n - 1, \max \leftarrow X[n]$
M2: If $k == 0$ then exit
M3: If $X[k] \leq \max$ go to M5
M4: Set $j \leftarrow k, \max \leftarrow X[k]$
M5: Decrease k by one and go to M2
- Find the average case complexity of execution of M4. (5)
4. Given a sequence of n matrices $A_1, A_2, \dots, A_n$, and their dimensions $p_0, p_1, p_2, \dots, p_n$, where $i = 1, 2, \dots, n$, matrix A_i has dimension $p_{i-1} \times p_i$. You need find out $A_1 \times A_2 \times \dots \times A_n$. Propose an algorithm to determine (a) the minimum number of multiplications required, (b) the order of multiplication i.e., parenthesization of the matrices to achieve the minimum multiplication. c) Find complexity of your algorithm. (3+2+2)
5. Provide a sketch of steps (efficient of course!) for the following problems:
- (a) If S is a set of vertices in an undirected graph $G = (V, E)$, define $f(S)$ to be the length of the shortest edge between a vertex in S and a vertex not in S , i.e., $f(S) = \min\{(v, w) : v \in S, w \notin S, (v, w) \in E\}$. Present an efficient algorithm for the following problem. Input: a connected, undirected graph G , with a non-negative length $l(e)$ on each edge e . Output: a non-empty set S that makes $f(S)$ as large as possible, subject to the requirement that $S \neq V$. (2)
- (b) Given a directed acyclic graph in which there is exactly one source node s and one sink node t . Give an efficient algorithm to find out the number of paths between s and t . (3)
- (c) Consider the polynomial $P = x + 2x^2 + 3x^3 + 4x^4 + 5x^5 + 6x^6 + 7x^7$. We represent such polynomial as follows $P_{0,1,\dots,7}$ and $P_{0,1,\dots,7}(1, \omega, \dots, \omega^7)$ denotes the value of the polynomial at $x = 1, \omega, \omega^2, \dots, \omega^7$. We use fft/dft strategy to determine the same. Therefore, we need to calculate $P_{0,2,4,6}(1, \omega^2, \omega^4, \omega^6)$ and $P_{1,3,5,7}(1, \omega^2, \omega^4, \omega^6)$ and these can be computed recursively. Using fft/dft techniques, find the value of $P_{0,2,4,6}(1, \omega^2, \omega^4, \omega^6)$ only. Please show all steps. (3)
- (d) The square of a matrix A is its product with itself, AA . Show that five multiplications are sufficient to compute the square of a 2×2 matrix. (2)